Package 'primace'

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Title Prediction Tool for One year MACE following Primary PCI

Version 0.0.0.9000

Description Prediting possibility of first year major adverse cardiovascular events (MACE) following primary percutaneous coronary intervention (PPCI).

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Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.1.2

StagedInstall no

Suggests testthat (>= 3.0.0)

Config/testthat/edition 3

Imports shinydashboard (>= 0.7.2), shiny (>= 1.7.1), shinybusy (>= 0.3.1), readxl (>= 1.3.1), randomForestSRC (>= 3.1.1), pROC (>= 1.18.0), pracma (>= 2.4.2), mlr3proba (>= 0.4.13), mlr3extralearners (>= 0.5.49-9000), mlr3learners (>= 0.5.4), mlr3 (>= 0.14.1), ggplot2 (>= 3.3.5), foreign (>= 0.8-81), dplyr (>= 1.0.7), caret (>= 6.0-89), data.table (>= 1.14.6), devtools (>= 2.4.3), rsconnect (>= 0.8.25), readr (>= 2.0.2), writexl, haven, forcats (>= 0.5.2), shinydashboardPlus (>= 2.0.3), mlr3tuning, here

Depends R (>= 3.5.0)

LazyData true

Collate 'data.R' 'utilities.R' 'primace_init.R' 'styles.R' 'global.R' 'analytics.R' 'server.R' 'ui.R' 'best_point.R' 'final_predict.R' 'load_model.R' 'loading_function.R' 'manual_prediction.R' 'max_perf_calc.R' 'predict_scores.R' 'preprocessing.R' 'reactiveVal_output.R' 'survROC.R' 'upload_file.R'

NeedsCompilation no

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`best_point` *Best Point Finder*

Description

Finding best point of ROC curve according to youden and closest topleft method.

Usage

```
best_point(c, se, sp)
```

Arguments

`c`	c_rank scores
`se`	sensitivity
`sp`	specificity

Value

A list containing youden index and closest topleft points

complete_case *Complete Case Dataset*

Description

creating a dataset with no missing values

Usage

```
complete_case(low_miss_dataframe)
```

Arguments

low_miss_dataframe	returned dataset with less than ten percent missing values from check_missing function, ie. missing_data function third value

Value

a list containing complete case dataset and included variables in complete case

final_predict *Final Prediction*

Description

Prediction of each observation according to best points found by youden and closest topleft method.

Usage

```
final_predict(predict_table, best_scores, target)
```

Arguments

predict_table	predicted table of crank scores of observations
best_scores	result of survROC function
target	whether time and status variables are available

Value

predicted table according to best cut-off points of youden index and closest top left point

getmode *get mode of a vector*

Description

to find mode of a vector of items

Usage

```
getmode(v)
```

Arguments

v	vetor of items

Value

mode of vector

load_dependencies *Packages loader*

Description

Loading packages and sources

Usage

```
load_dependencies()
```

Details

Not Run in shiny cloud.

Value

None

load_model *Model Loader Loading trained models on original study*

Description

Model Loader Loading trained models on original study

Usage

```
load_model(
  input,
  rf_model = "models/rfsrc.learner_mod.RDS",
  xgb_model = "models/xgboost.learner_mod.RDS"
)
```

Arguments

input	input of shinyserver
rf_model	path to survival random forest model
xgb_model	path to xgboost model

Value

loaded models in rds format

max_perf_calc *Performance Calculator creating confusion matrix to show best performance result.*

Description

Performance Calculator creating confusion matrix to show best performance result.

Usage

```
max_perf_calc(predict_table)
```

Arguments

predict_table	predicted table of outcomes of each observation

Value

table of confusion matrix as dataframe

`predict_scores` *Score Predictor. Main predicting tool of data by saved pre-trained machine learning models*

Description

Score Predictor. Main predicting tool of data by saved pre-trained machine learning models

Usage

```
predict_scores(data, model)
```

Arguments

`data`	provided dataset we want to predict
`model`	saved model to be used as prediction model

Value

predited scores table as datatable.

`preprocessing` *Data Preprocessing*

Description

module to perform preprocessing on dataset

Usage

```
preprocessing()
```

Value

uploaded preprocessed dataset

`primace_init` *Runner file initializing app*

Description

Runner file initializing app

Usage

```
primace_init()
```

raw_variables *raw dataset variable names*

Description

raw dataset variable names

Usage

```
data(test_set)
```

Format

A vector of raw dataset variables names

Author(s)

Behnam Hedayat <dr.hedayatb@gmail.com>

Source

source dataset

References

primace.aikadeh.com

survROC *survROC. modified function of survivalROC package to return specificity as well. It is used for prediction of survival according to time of interest.*

Description

survROC. modified function of survivalROC package to return specificity as well. It is used for prediction of survival according to time of interest.

Usage

```
survROC(
  Stime,
  status,
  marker,
  entry = NULL,
  predict.time,
  cut.values = NULL,
```

```
  method = "NNE",
  lambda = NULL,
  span = NULL,
  window = "symmetric"
)
```

Arguments

Stime	event or censor times
status	status at event or censor time
marker	crank scores
entry	entry of survial time
predict.time	time to predict
cut.values	predefined cutoff of crank scores
method	method to make analysis, either NNE or KM
lambda	defining lambda value
span	defining span value for NNE method
window	defining window

Value

A list containing cut-off values, true positive rate, false positive rate, specificity, prediction time, area under the curve at specified prediction time, cumulative survival.

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test_set *test_set Dataset*

Description

test_set Dataset

Usage

```
data(test_set)
```

Format

A data frame containig example test set of main dataset

Author(s)

Behnam Hedayat <dr.hedayatb@gmail.com>

Source

source dataset

References

primace.aikadeh.com

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varnames *varnames Dataset*

Description

varnames Dataset

Usage

```
data(varnames)
```

Format

A data frame containig variable names with their respective types and levels

Author(s)

Behnam Hedayat <dr.hedayatb@gmail.com>

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